NASPIER IMU BOARD V2

NASPIER IMU FFC

NASPIER CORE

NASPIER CARRIER

SENSOR_BOARD

Dosya: SENSOR_IMU.kicad_sch

FLEX_CABLE

Dosya: FLEX_CABLE_sch.kicad_sch

CORE

Dosya: CORE.kicad_sch

CARRIER

Dosya: CARRIER.kicad_sch

Schematic	Naspier Core	Create at	2026-09-20
		Update at	2026-10-29
Board	NASPIER CORE FLIGHT CONTROLLER SYSTEM	Page	SYSTEM-EXPOSE
Drawn	Ahmet Vapurcu	Naspier Core	
Reviewed			
		Version	Size / Page 1 Total 16
		REV0 VER0	A3 / VAP-SAN

HIROSE 34P
FFC CONNECTOR

CORE_H743

STM32H743IIK6

ARM® Cortex®-M7 STM32H7 Microcontroller IC
32-Bit 480MHz 2MB (2M x 8) FLASH
201 UFBGA (10x10)

Dosya: CORE_H743.kicad_sch

CORE_POWER

IMU BUSSES 3.3V
REGULATORS

FC & AUX 3.3V
REGULATORS

Dosya: CORE_POWER.kicad_sch

CORE_ONBOARD

AT24C64D
EEPROM-I2C

FRAM
SPI

ICM-45686
IMU-SPI

MS5611
BARO-I2C

Dosya: CORE_ONBOARD.kicad_sch

CORE_CONNECTORS

HIROSE 100P
CONNECTOR

HIROSE 50P
CONNECTOR

Dosya: CORE_CONNECTORS.kicad_sch

Schematic	Naspier Core	Create at	2026-09-20
		Update at	2026-10-29
Board	NASPIER CORE FLIGHT CONTROLLER SYSTEM	Page	CORE BOARD
Drawn	Ahmet Vapurcu	Naspier Core	
Reviewed			
		Version	Size / Page 2 Total 16
		REVO VERO	A4 / VAP-SAN

U1A
STM32H743IIK6

U1B
STM32H743IIK6

ARM® Cortex®-M7 STM32H7 Microcontroller IC
32-Bit 480MHz 2MB (2M x 8) FLASH
201 UFBGA (10x10)
NON-STANDARD PIXHAWK CHIP

FMU 3V3 DIGITAL LINE CAPS

FMU 3V3 ANALOG LINE CAPS

R-G-B LEDS

CRYSTALS

SPI4 EXPANSION

REV SENS DO NOT PLACE

RTC BATTERY LINE

C300 DNP: VBAT decoupling for testing only.
MLCC leakage drains the coin cell.

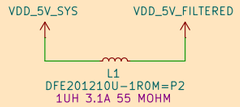
DEBUG INTERFACE

THIS CONNECTOR JUST FOR DEBUG/PROGRAMMING POINTS GROUPS

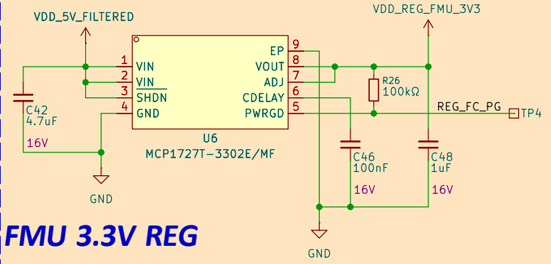
PWR_FLAG (ERC drivers for the rails on this sheet)

Schematic	Naspier Core		Create at	2026-09-20
Board	NASPIER CORE FLIGHT CONTROLLER SYSTEM		Update at	2026-10-29
Drawn	Ahmet Vapurcu		Page	CORE_H743
Reviewed		Naspier Core		
		Version: REV0 VER0	Size: A3	Page 3 Total 16
				VAP-SAN

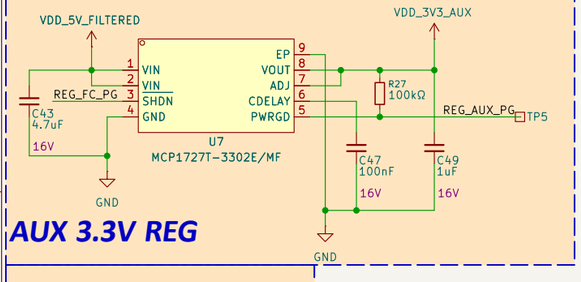
5V LINE FILTER



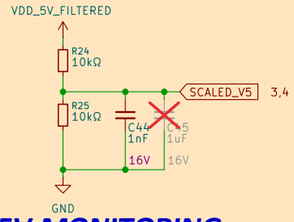
FMU 3.3V REG



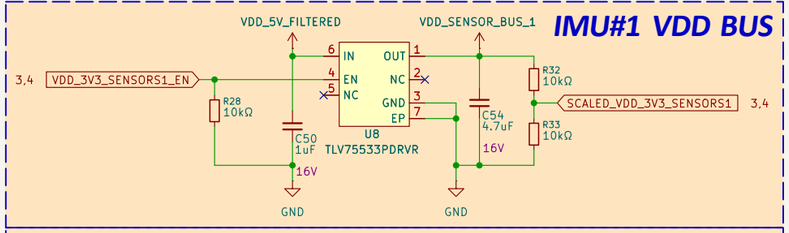
AUX 3.3V REG



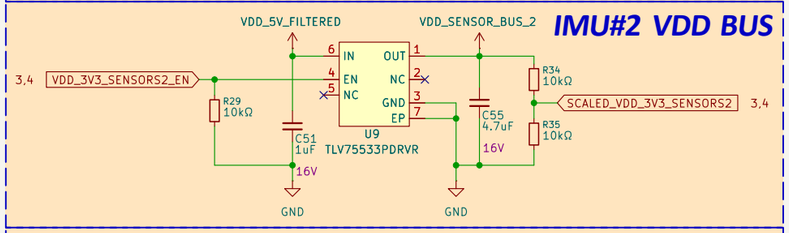
5V MONITORING



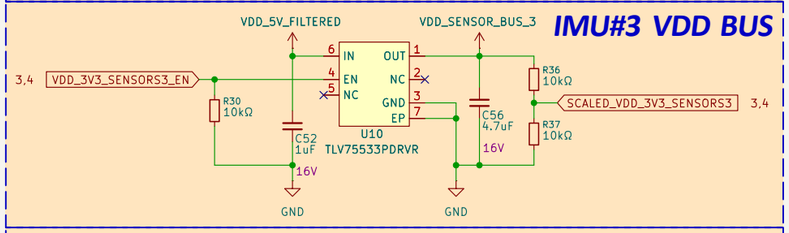
IMU#1 VDD BUS



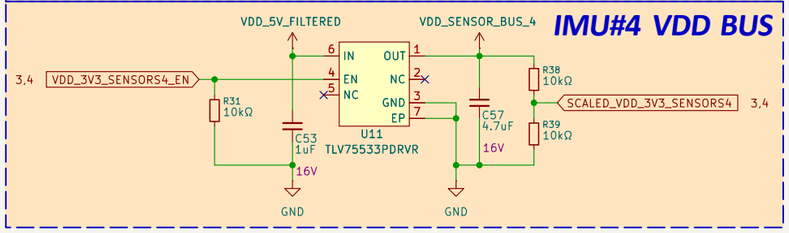
IMU#2 VDD BUS



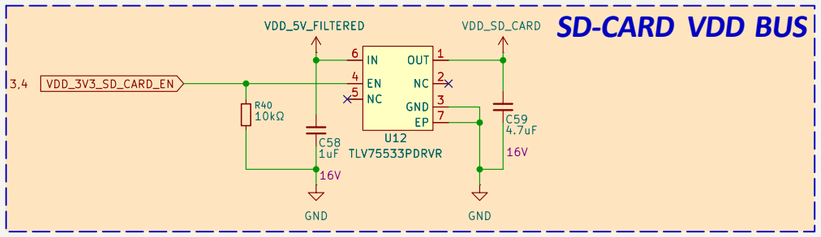
IMU#3 VDD BUS



IMU#4 VDD BUS

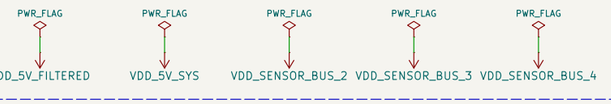


SD-CARD VDD BUS



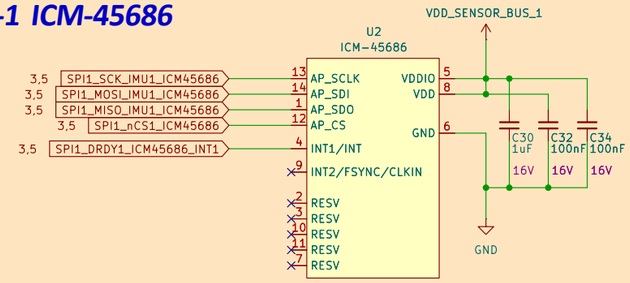
NOT APPROVED (B-LDO, 2026-10-25): LDL212PV33R -> TLV75533PDRVR, WSON-6 2x2 (TI DRV).
Pins: 1 OUT, 2 NC, 3 GND, 4 EN, 5 NC, 6 IN, 7 pad. Pins 2 / 5 float (no-connect flags).
VIN 5.5 V recommended, 6 V absolute max (LDL212: 18 V): 5 V rail spikes (hot plug, servo back-feed) can exceed it. To check later: TVS or LC filter at IN if needed. 500 mA max, no sense pin.

PWR_FLAG (ERC drivers for the rails on this sheet)

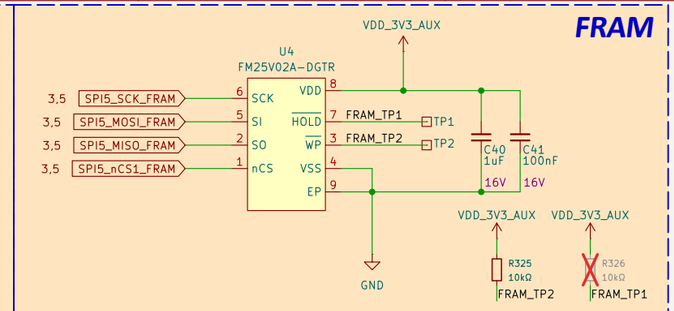


Schematic	**Naspier Core**	Create at	2026-09-20
		Update at	2026-10-29
Board	NASPIER CORE FLIGHT CONTROLLER SYSTEM	Page	**CORE_POWER**
Drawn	Ahmet Vapurcu	**Naspier Core**	
Reviewed			
		Version	Size / Page 4 Total 16
		REV0 VER0	A3 / VAP-SAN

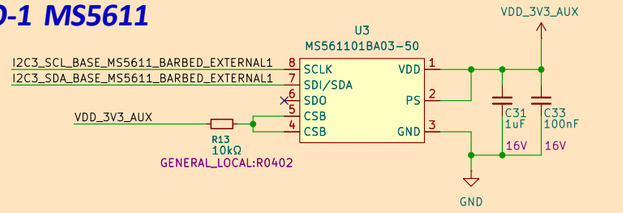
IMU-1 ICM-45686



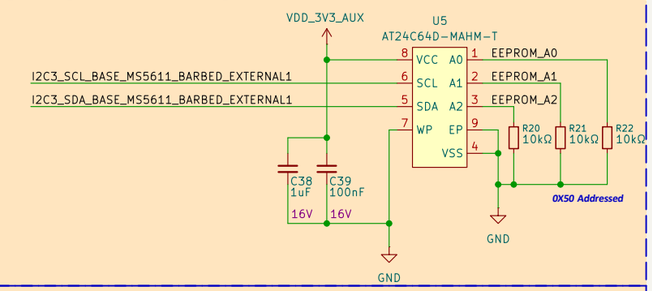
FRAM



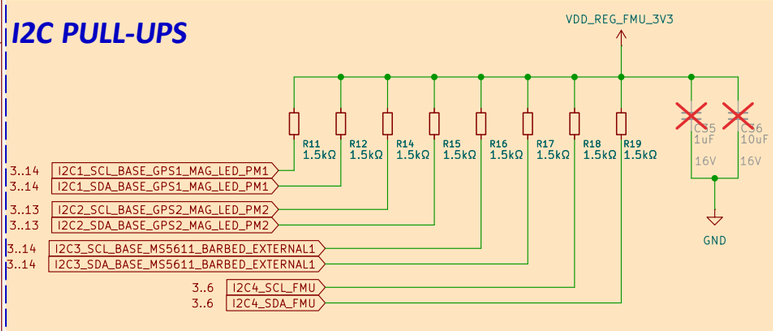
BARO-1 MS5611



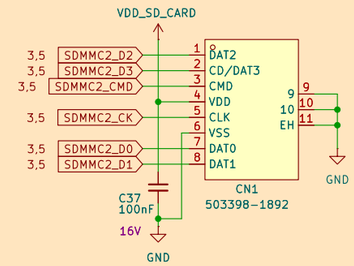
ONBOARD EEPROM



I2C PULL-UPS

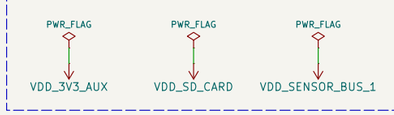


SD-CARD



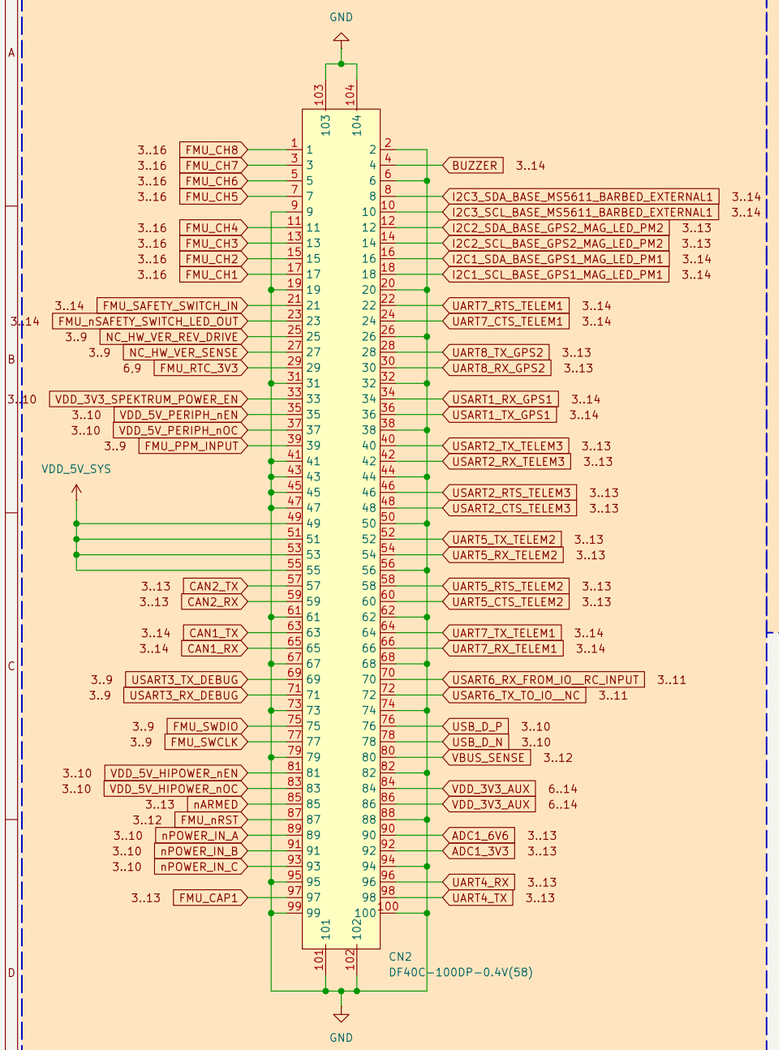
Push-In
Push-Out

PWR_FLAG (ERC drivers for the rails on this sheet)



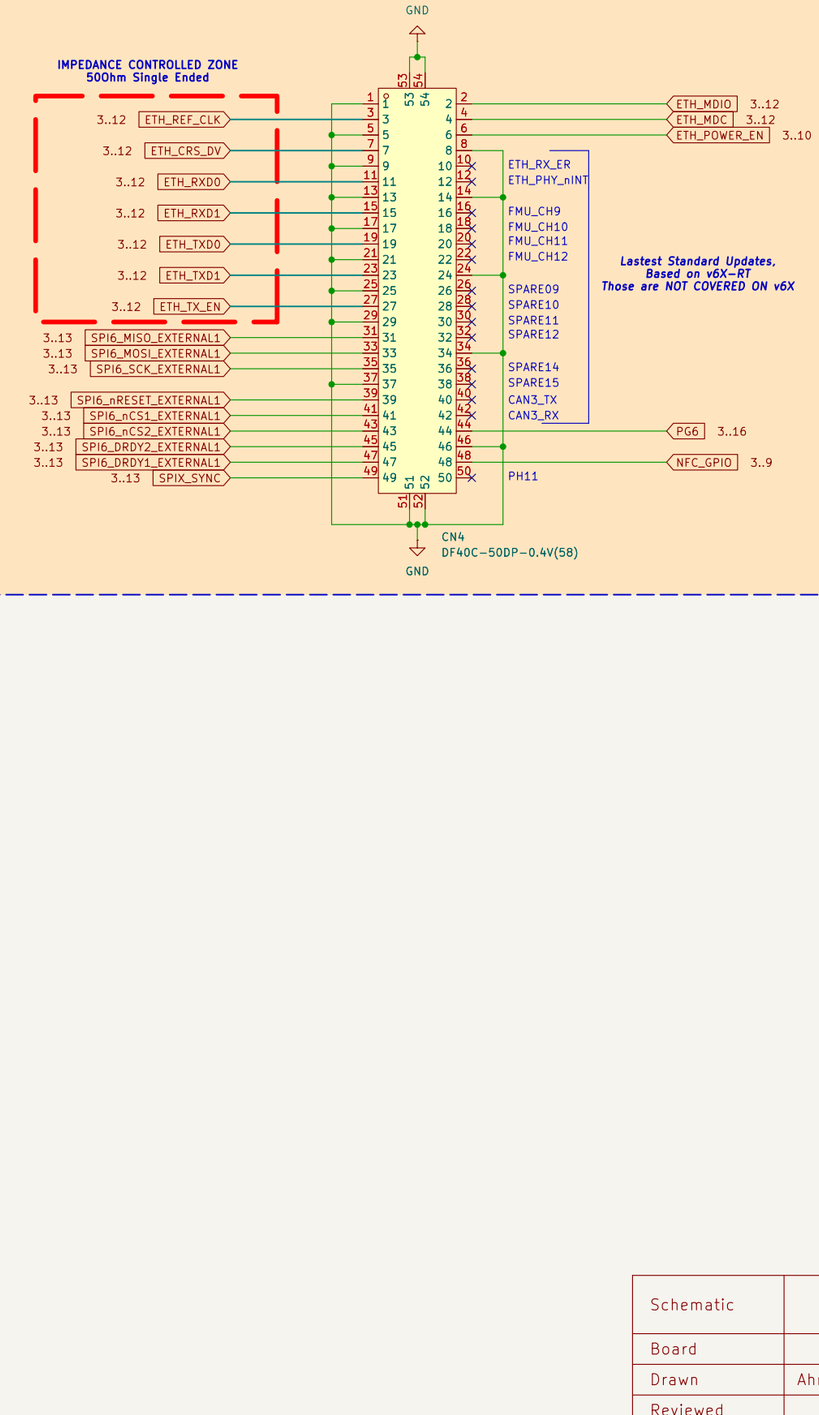
Schematic	**Naspier Core**	Create at	2026-09-20
		Update at	2026-10-29
Board	NASPIER CORE FLIGHT CONTROLLER SYSTEM	Page	**CORE_ELEMENTS**
Drawn	Ahmet Vapurcu	**Naspier Core**	
Reviewed			
		Version	Size / Page 5 Total 16
		REVO VERO	A3 / VAP-SAN

PAB X1 100P HIROSE

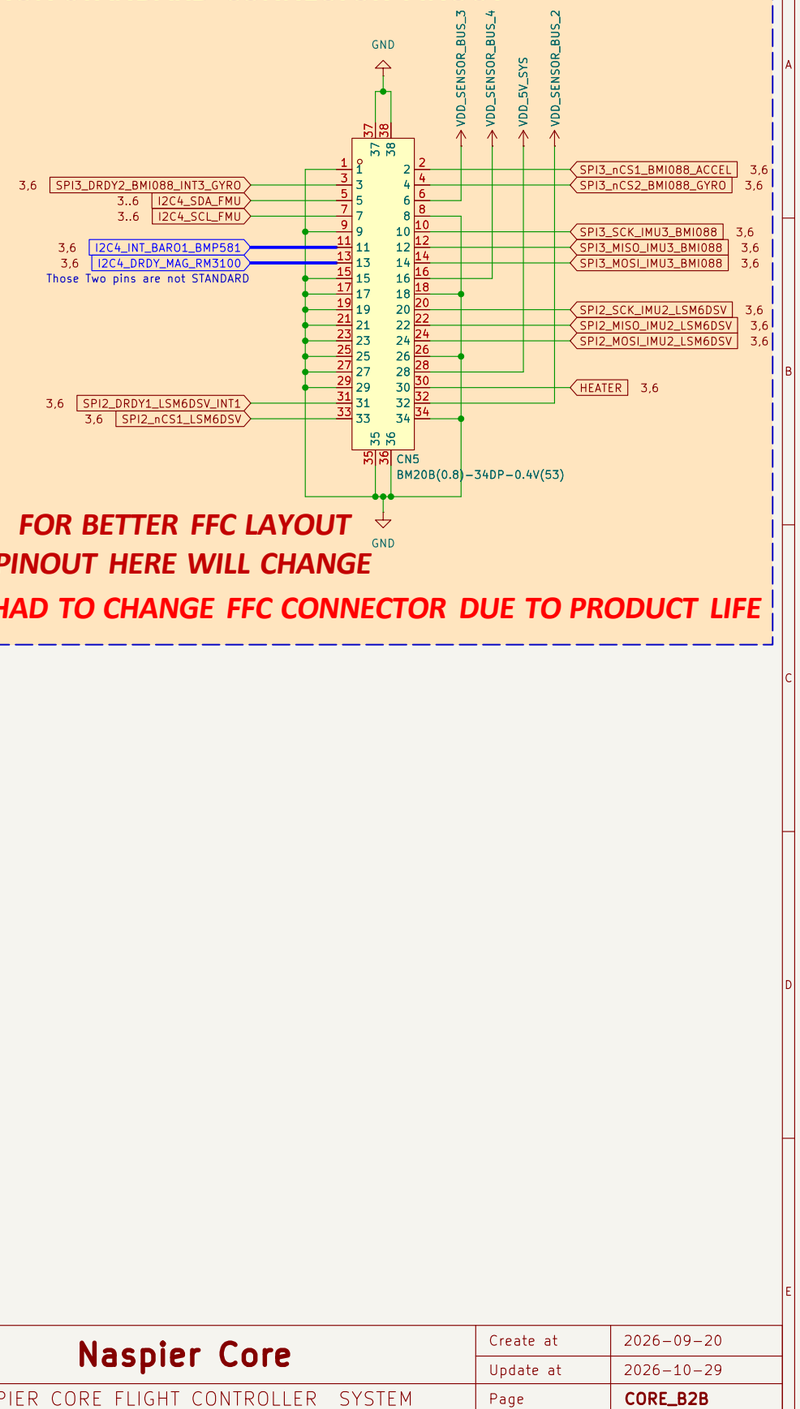


PAB X2 50P HIROSE

v6X-RT only (not on v6X cores): FMU_CH9-CH12, CAN3, ETH_RX_ER, ETH_PHY_nINT, SPARE09-12 / 14 / 15, PH11. Fit / route them only for v6X-RT support.



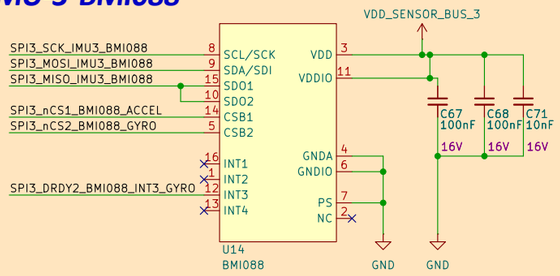
PAB X3 34P SENSOR FFC CONNECTOR
NON-STANDARD CONNECTOR PINOUT



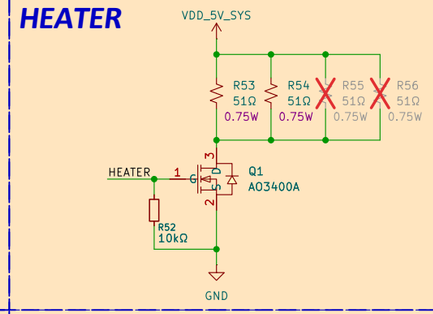
FOR BETTER FFC LAYOUT
PINOUT HERE WILL CHANGE
HAD TO CHANGE FFC CONNECTOR DUE TO PRODUCT LIFE

Schematic	**Naspier Core**	Create at	2026-09-20
		Update at	2026-10-29
Board	NASPIER CORE FLIGHT CONTROLLER SYSTEM	Page	**CORE_B2B**
Drawn	Ahmet Vapurcu	**Naspier Core**	
Reviewed			
		Version	Size	Page 6 Total 16
		REV0 VER0	A3	VAP-SAN

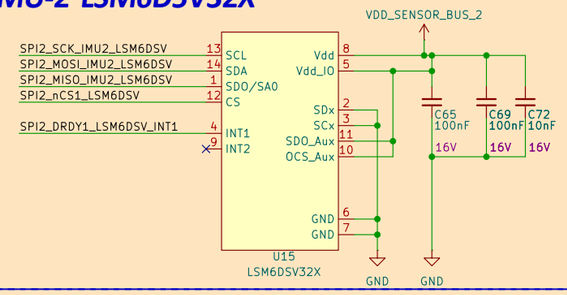
IMU-3 BMI088



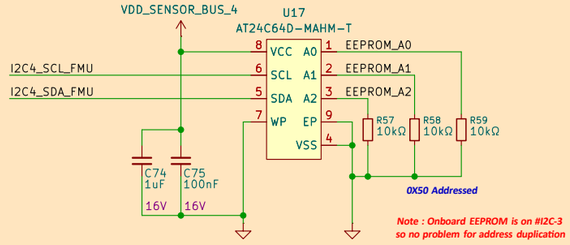
HEATER



IMU-2 LSM6D5V32X

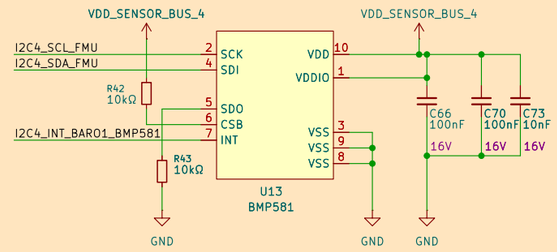


EEPROM

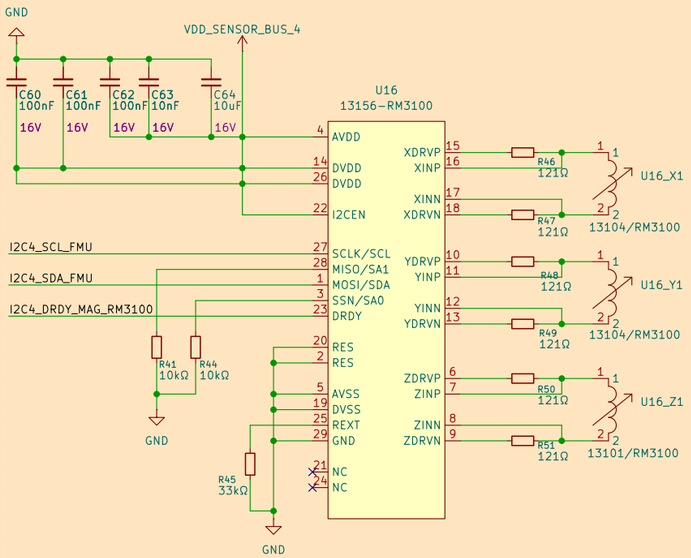


C11: R53–R56 Rohm LTR18EZPF51R0, 0.75 W wide-terminal 1206 (0612 footprint).
R55 / R56 DNP: fit them for more heater power when needed.

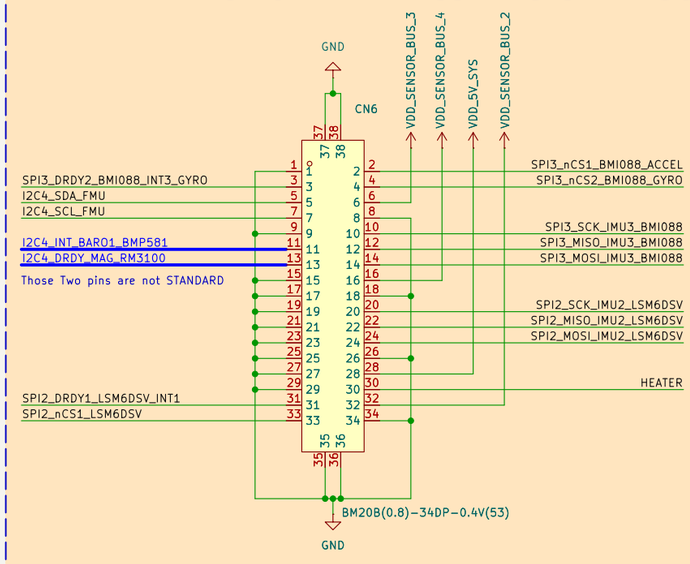
BARO-2 BMP581



MAG-RM3100



PAB X3 34P SENSOR FFC CONNECTOR
NON-STANDARD CONNECTOR PINOUT



FOR BETTER FFC LAYOUT
PINOUT HERE WILL CHANGE

Schematic	**Naspier Core**	Create at	2026-09-20
Board	NASPIER CORE FLIGHT CONTROLLER SYSTEM	Update at	2026-10-29
Drawn	Ahmet Vapurcu	Page	**SENSOR_BOARD**
Reviewed		**Naspier Core**	
		Version: REVO VERO	Size: A3
		Page 7 Total 16	VAP-SAN

CARRIER_B2B



Dosya: CARRIER_B2B.kicad_sch

CARRIER_CONN_1



Dosya: CARRIER_CONN_1.kicad_sch

CARRIER_IO_MCU



Dosya: CARRIER_IO_MCU.kicad_sch

CARRIER_POWER



Dosya: CARRIER_POWER.kicad_sch

CARRIER_CONN_2



Dosya: CARRIER_CONN_2.kicad_sch

CARRIER_ELEMENTS



Dosya: CARRIER_ELEMENTS.kicad_sch

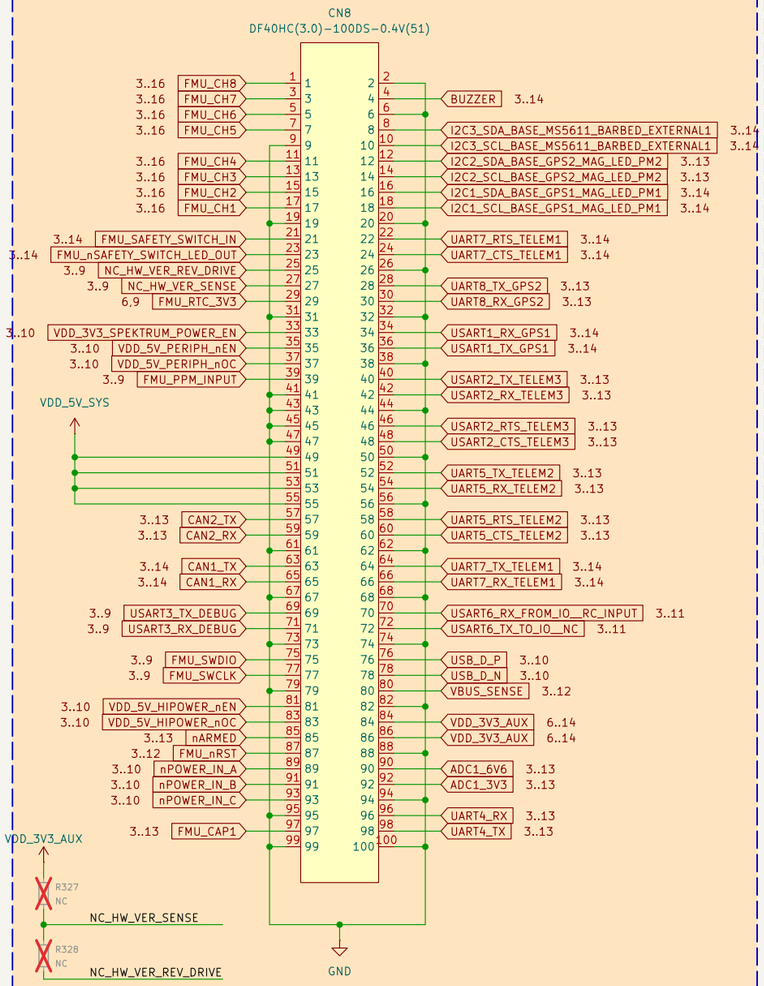
CARRIER_PWM



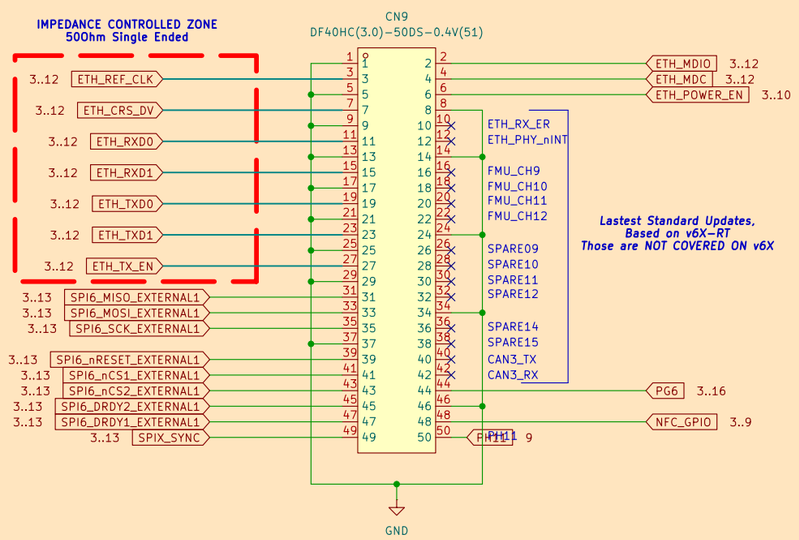
Dosya: CARRIER_PWM.kicad_sch

Schematic	**Naspier Core**	Create at	2026-09-20
		Update at	2026-10-29
Board	NASPIER CORE FLIGHT CONTROLLER SYSTEM	Page	**CARRIER BOARD**
Drawn	Ahmet Vapurcu	**Naspier Core**	
Reviewed			
		Version	Size	Page 8 Total 16
		REVO VERO	A4	VAP-SAN

PAB X1 100P HIROSE



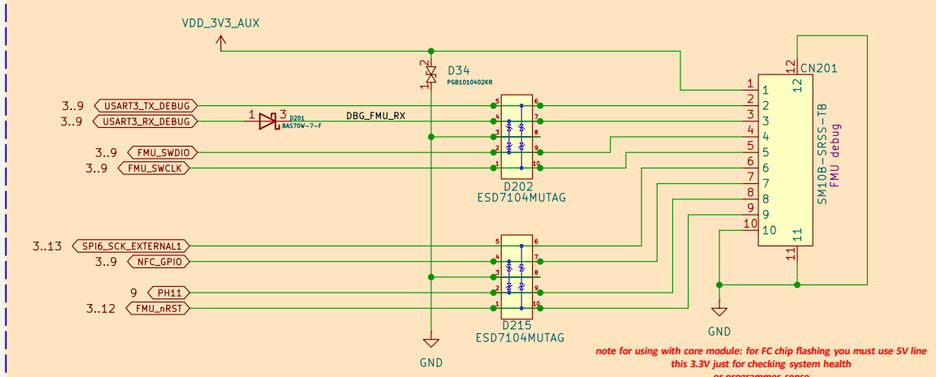
PAB X2 50P HIROSE



Lastest Standard Updates,
Based on v6X-RT
Those are NOT COVERED ON v6X

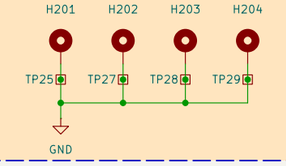
v6X-RT only (not on v6X cores): FMU_CH9–CH12, CAN3, ETH_RX_ER, ETH_PHY_nINT, SPARE09–12 / 14 / 15, PH11. Fit / route them only for v6X-RT support.

FMU DEBUG



note for using with core module: for FC chip flashing you must use 5V line
this 3.3V just for checking system health
or programmer sense
Other Standard module's fc chip can be powered by this bus

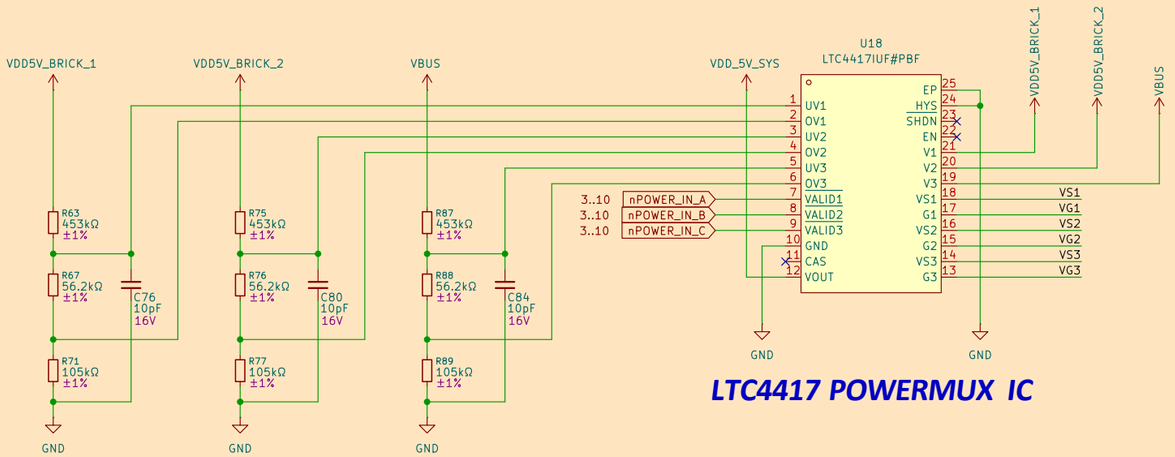
MountingHole M2 x4
CORE SCREWED STANDOFFS



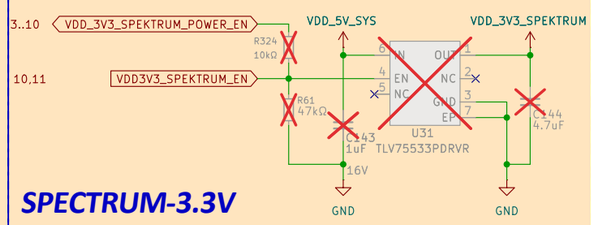
HW_VER_SENSE divider (RC05 page 2, R13 / R18): R327 from FMU_VDD_3V3 (X1 84 / 86 = VDD_3V3_AUX), R328 to HW_VER_REV_DRIVE. Both DNP; values set the baseboard version. Net names keep the NC_ prefix.

Schematic	**Naspier Core**	Create at	2026-09-20
		Update at	2026-10-29
Board	NASPIER CORE FLIGHT CONTROLLER SYSTEM	Page	**CARRIER_B2B**
Drawn	Ahmet Vapurcu	**Naspier Core**	
Reviewed			
		Version	Size / Page 9 Total 16
		REVO VERO	A3 / VAP-SAN

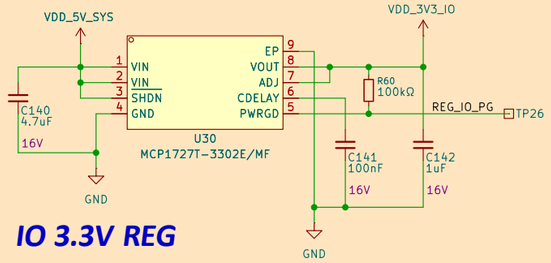
LTC4417 POWERMUX IC



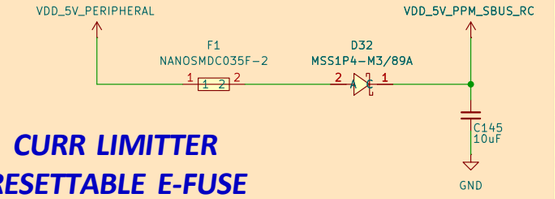
SPECTRUM-3.3V



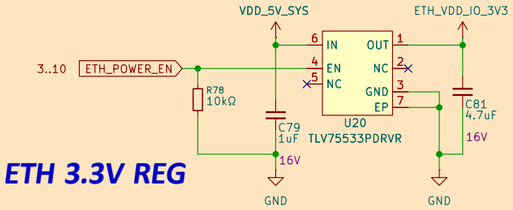
IO 3.3V REG



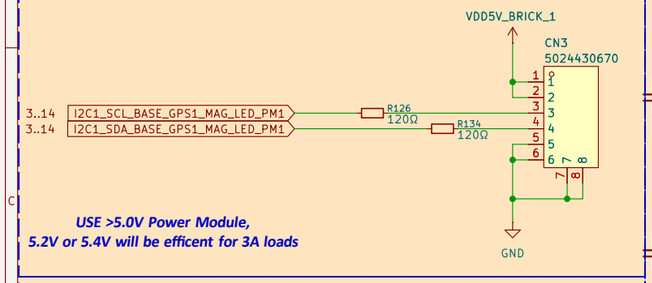
CURR LIMITTER RESETTABLE E-FUSE



ETH 3.3V REG

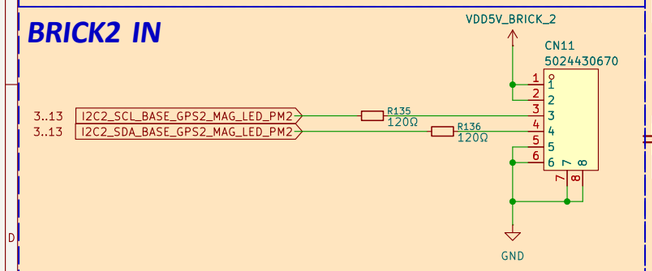


BRICK1 IN

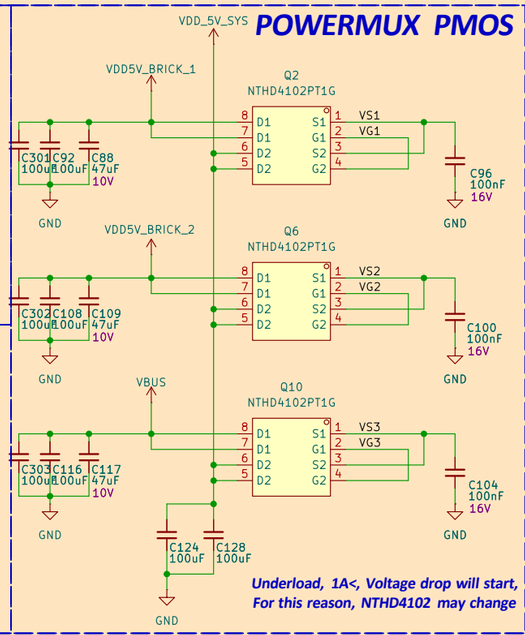


USE >5.0V Power Module,
5.2V or 5.4V will be efficent for 3A loads

BRICK2 IN

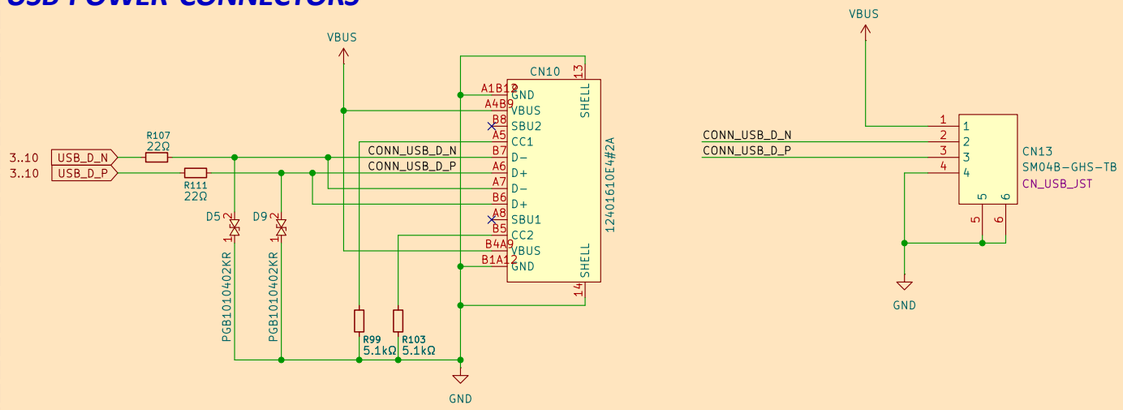


POWERMUX PMOS



Underload, 1A<, Voltage drop will start,
For this reason, NTHD4102 may change

USB POWER CONNECTORS

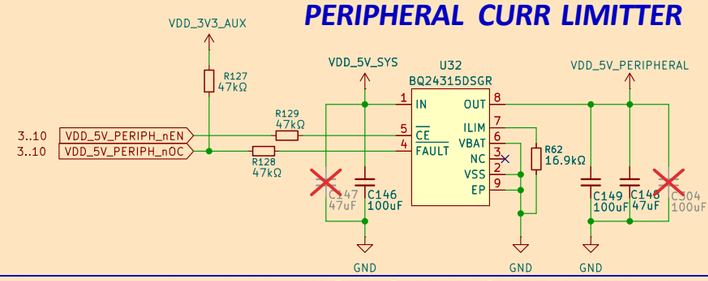


PWR_FLAG (ERC drivers for the rails on this sheet)

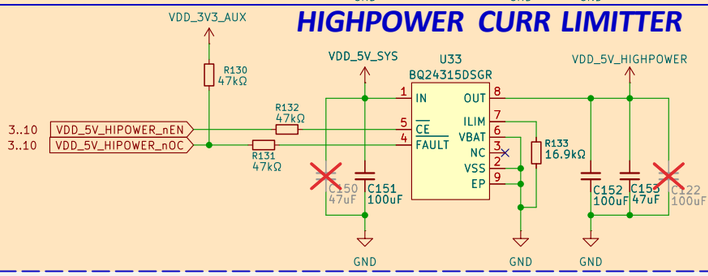
PWR_FLAG VBUS
PWR_FLAG VDD5V_BRICK_1
PWR_FLAG VDD5V_BRICK_2
PWR_FLAG VDD_3V3_SPEKTRUM

TP30 VDD_5V_SYS
TP31 VDD_3V3_AUX
TP32 VDD_3V3_IO
TP33 VDD_5V_PERIPHERAL
TP34 VDD_5V_HIGHPOWER
TP35 GND

PERIPHERAL CURR LIMITTER



HIGHPOWER CURR LIMITTER

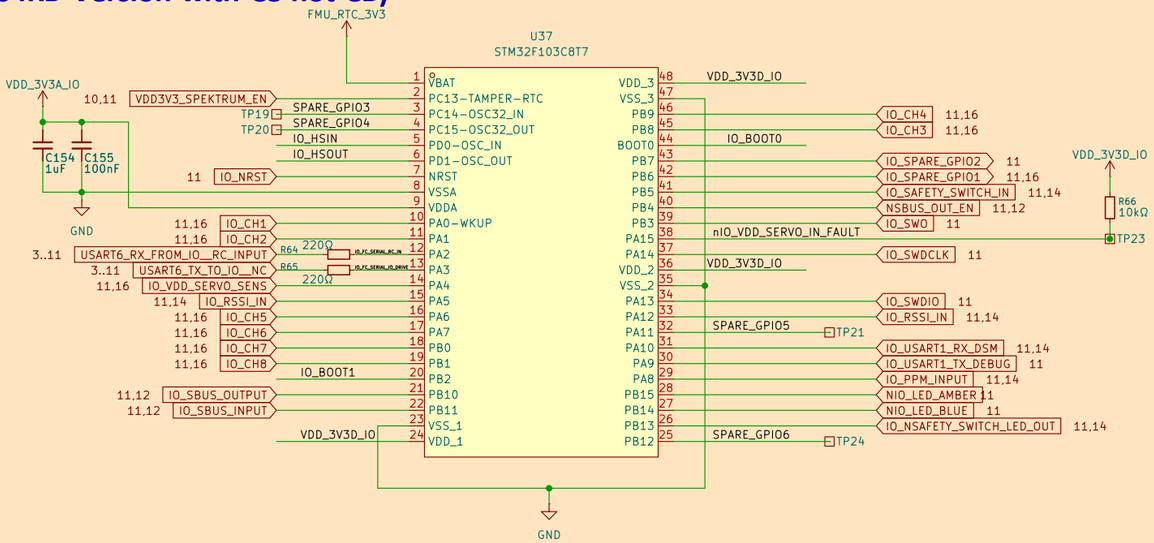


NOT APPROVED (B-LDO, 2026-10-25): LDL212PV33R -> TLV75533PDRVR, WSON-6 2x2 (TI DRV).
Pins: 1 OUT, 2 NC, 3 GND, 4 EN, 5 NC, 6 IN, 7 pad. Pins 2 / 5 float (no-connect flags).
VIN 5.5 V recommended, 6 V absolute max (LDL212: 18 V): 5 V rail spikes (hot plug, servo back-feed) can exceed it. To check later: TVS or LC filter at IN if needed. 500 mA max, no sense pin.

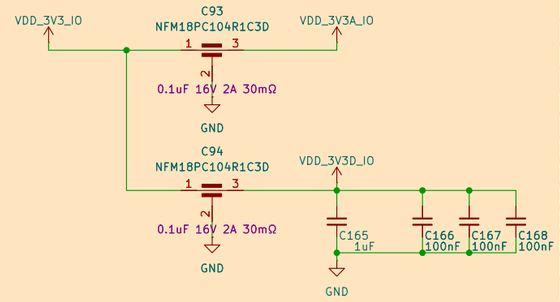
L6 (RC05 page 8): U31 EN = VDD3V3_SPEKTRUM_EN, driven by the IO MCU (PC13) directly and by the FMU (PH2, VDD_3V3_SPEKTRUM_POWER_EN, DS-012) through R324 10 k; R61 47 k pull-down. Both are the regulator enable (the receiver has no chip enable): the IO MCU can power-cycle it for DSM bind.

Schematic	**Naspier Core**	Create at	2026-09-20
		Update at	2026-10-29
Board	NASPIER CORE FLIGHT CONTROLLER SYSTEM	Page	**CARRIER_POWER**
Drawn	Ahmet Vapurcu	**Naspier Core**	
Reviewed			
		Version	Size / Page 10 Total 16
		REV0 VER0	A3 / VAP-SAN

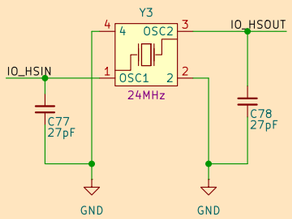
IO F103 CHIP
(64KB Version with C8 not CB)



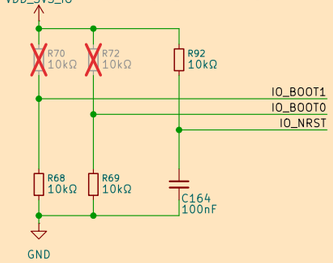
IO POWER FILTER

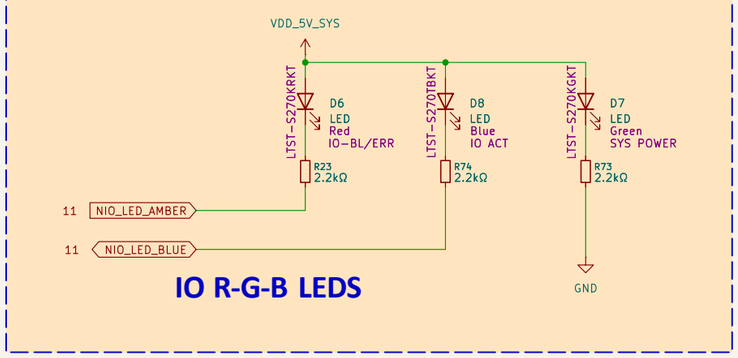


IO CRYSTAL



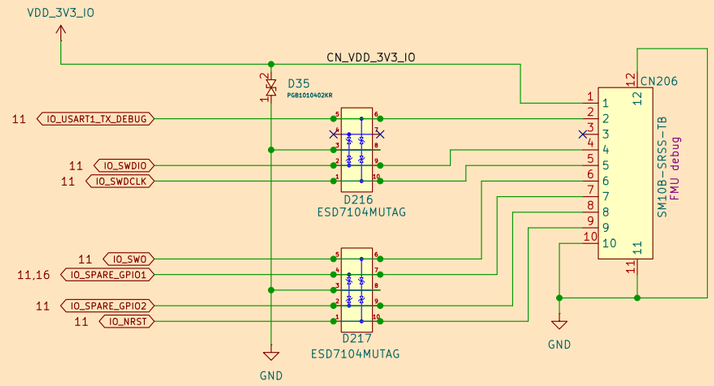
IO BOOT-NRST

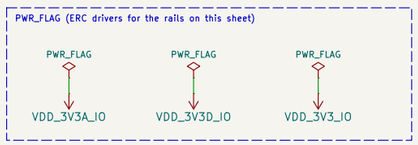




IO R-G-B LEDS

IO DEBUG





Schematic	**Naspier Core**	Create at	2026-09-20
		Update at	2026-10-29
Board	NASPIER CORE FLIGHT CONTROLLER SYSTEM	Page	**CARRIER_IO_MCU**
Drawn	Ahmet Vapurcu	**Naspier Core**	
Reviewed			
		Version	Size	Page 11 Total 16
		REVO VERO	A3	VAP-SAN

CHECK (C17): RC05 drives ETH_NRESET from the MIC5332 POR output.
Here nRST has only R116 (10 k pull-up): the PHY relies on its internal POR.

v6X-RT only: ETH_RX_ER (X2 pin 10) and R112 are used with v6X-RT cores, not with v6X.

ETH_VDD_IO_3V3

C159 10uF, C158 100nF, C157 100nF

C160 100nF, C161 10nF

R114 1.5kΩ, R116 10kΩ

ETH_MDIO
ETH_NRST

C162 1uF, C163 470pF

ETH_VDDCR

U19
LAN8742AI-CZ

1	VDD2A	RXN	22
19	VDD1A	RXP	23
9	VDDIO	TXN	20
6	VDDCR	TXP	21
10	RXER/PHYAD0	LED2/nINT/nPME/nINTSEL	2
11	CRS_DV/MODE2	LED1/nINT/nPME/REGOFF	3
12	MDIO	XTAL2	4 ETH_HSOUT
13	MDC	XTAL1/CLKIN	5 ETH_HSIN
16	TXEN	RBIAS	24
15	nRST	EPAD	25
18	TXD1		
17	TXD0		
7	RXD1/MODE1		
8	RXD0/MODE0		
14	nINT/REFCLKO		

3..12 ETH_CRS_DV — R113 22Ω — R112
3..12 ETH_MDIO
3..12 ETH_MDC
3..12 ETH_TX_EN
ETH_NRST
3..12 ETH_TXD1
3..12 ETH_TXD0
3..12 ETH_RXD1 — R117 22Ω
3..12 ETH_RXD0 — R118 22Ω
3..12 ETH_REF_CLK — R120 22Ω

R121 12.1kΩ

ETH_VDD_IO_3V3

C85 100nF, R85 49.9Ω %1, R86 49.9Ω %1, R90 49.9Ω %1, R91 49.9Ω %1, C86 100nF

RX_N
RX_P
TX_N
TX_P

D27 ESD7104MUTAG

L2 DLW43MH201XK2L
L3 DLW43MH201XK2L

C87 100nF ISOLATION, C89 100nF ISOLATION
C90 100nF ISOLATION, C91 100nF ISOLATION

PHY_RX_N
PHY_RX_P
PHY_TX_N
PHY_TX_P

CN7 SM04B-GHS-TB ETH

D11 LED Blue ETH Link/Act
D10 LED
LTST-S270TBKT
R84 10kΩ
R83 1kΩ
R82 1kΩ

OSC2, OSC1, 25MHz Y4
C82 27pF
C83 27pF

ETH_VDD_IO_3V3
D31 LED Red ETH 3V3 POWER
LTST-S270KRKT
R154 1kΩ

MountingHole M3 x4 BASEBOARD HOLES

H205 H206 H207 H208

H205–H208: M3 baseboard holes, plain (no standoff):
3.1 mm drill, 5.0 mm plated pad to GND.

CARRIER EEPROM EXT SENSOR CAL.

VDD_3V3_AUX

U21
AT24C64D-MAHM-T

8	VCC	A0	1 EEPROM_A0
6	SCL	A1	2 EEPROM_A1
5	SDA	A2	3 EEPROM_A2
7	WP	EP	9
		VSS	4

3..14 I2C3_SCL_BASE_MS5611_BARBED_EXTERNAL1
3..14 I2C3_SDA_BASE_MS5611_BARBED_EXTERNAL1

R153 10kΩ, R151 10kΩ, R152 10kΩ
C114 1uF 16V, C115 100nF 16V
0X51 Addressed

VBUS SENS CIRCUIT

VBUS
R115 10kΩ
TP18
12,13 FMU_BOOTLOADER
R119 10kΩ

U22
MC74VHC1G32DFT1G
U26
MC74VHC1G09DFT1G

1	INB	VCC	5
2	INA		
3	GND	OUTY	4

VDD_3V3_AUX
VDD_5V_SYS
R123 1kΩ
TP22
VBUS_SENSE 3..12
C132 100nF
C136 100nF

PWR_FLAG (ERC drivers for the rails on this sheet)
PWR_FLAG
ETH_VDD_IO_3V3

RC INVERT BUFFER

VDD_3V3_IO
C203 100nF

U201
74LVC2G240DP
INVERT BUFFER

1	1OE	VCC	8
2	1A	2OE	7 RC_OE2
3	2Y	1Y	6
4	GND	2A	5

12,14 RC_1A
12,14 RC_2Y
R203 220
NSBUS_OUT_EN 11,12
IO_SBUS_INPUT 11,12
IO_SBUS_OUTPUT 11,12
R204 10k

NRST BUFFER

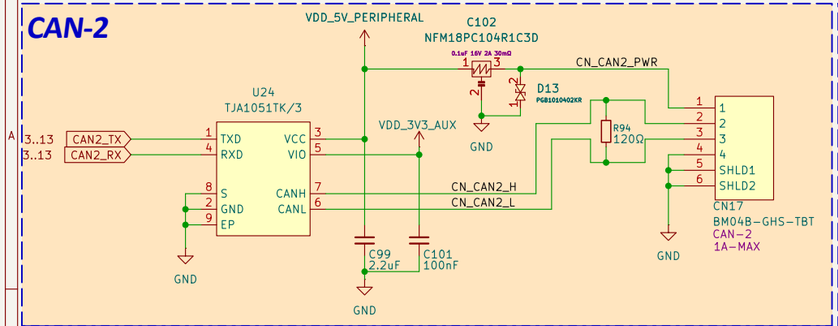
VDD_3V3_AUX
C208 100nF

U202
74LVC1G38
(NAND, open drain)

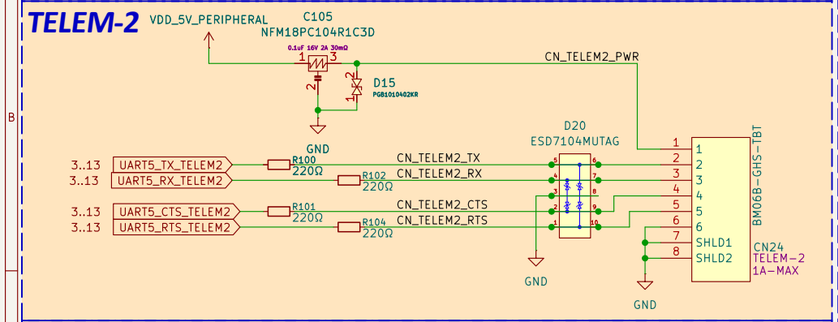
3..13 nARMED
12,13 FMU_RST_REQ
FMU_nRST 3..12

Schematic	Naspier Core	Create at	2026-09-20
		Update at	2026-10-29
Board	NASPIER CORE FLIGHT CONTROLLER SYSTEM	Page	CARRIER_ELEMENTS
Drawn	Ahmet Vapurcu	Naspier Core	
Reviewed			
		Version: REVO VERO	Size: A3
		Page 12 Total 16	VAP-SAN

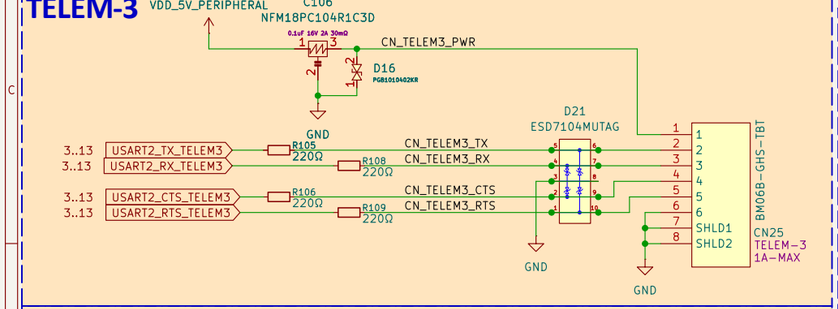
CAN-2



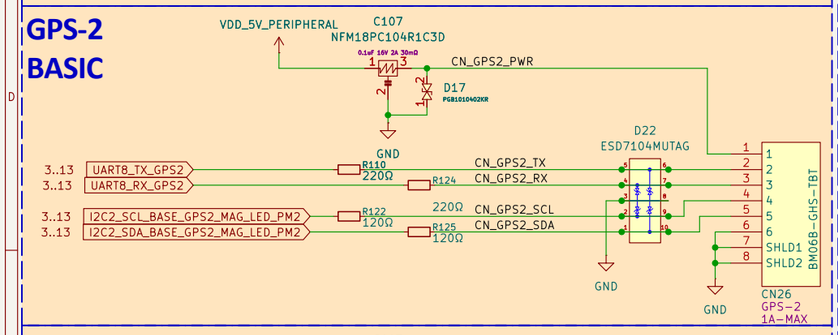
TELEM-2



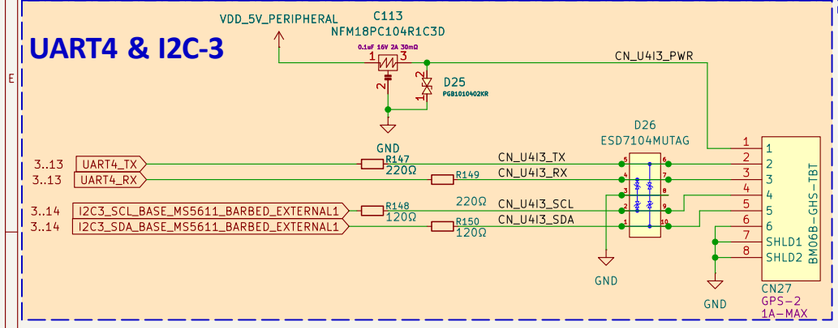
TELEM-3



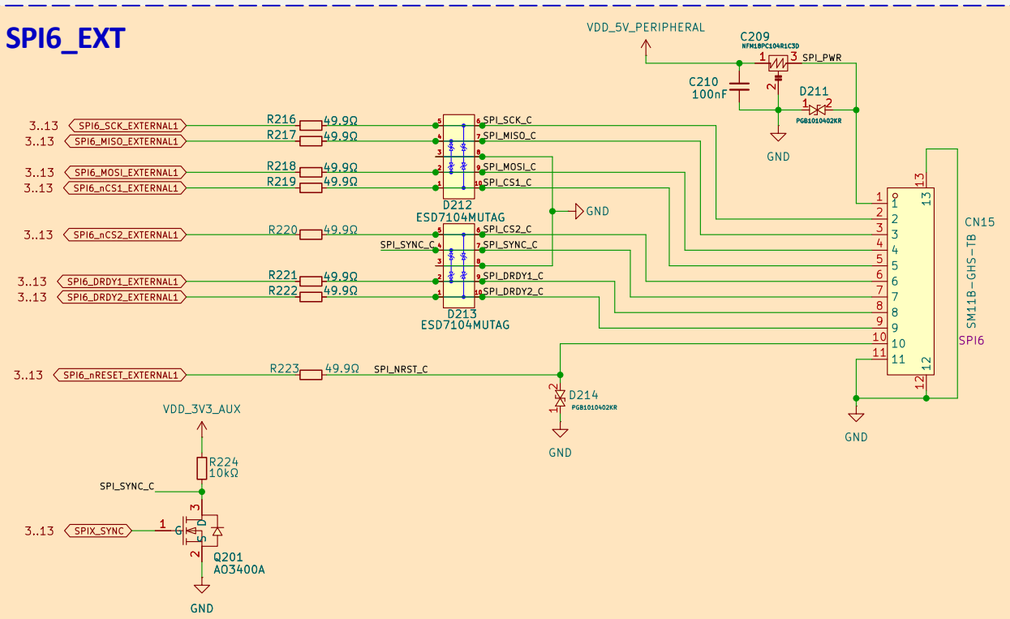
GPS-2 BASIC



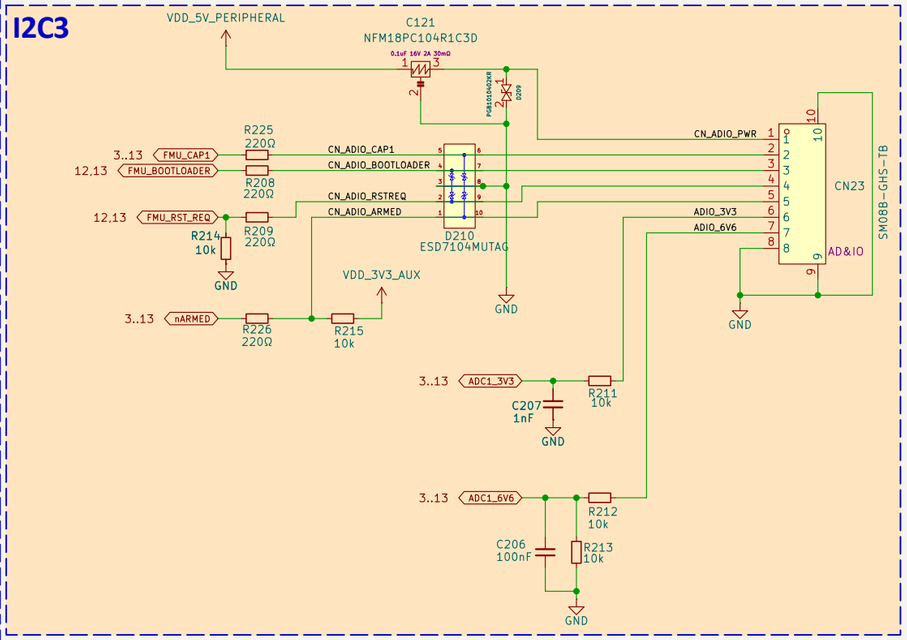
UART4 & I2C-3



SPI6_EXT



I2C3



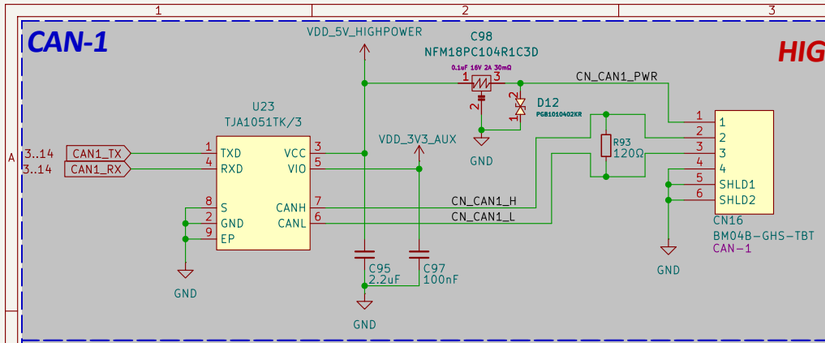
PWR_FLAG (ERC drivers for the rails on this sheet)



Schematic	**Naspier Core**	Create at	2026-09-20
		Update at	2026-10-29
Board	NASPIER CORE FLIGHT CONTROLLER SYSTEM	Page	**CARRIER_CONN_1**
Drawn	Ahmet Vapurcu	**Naspier Core**	
Reviewed			
		Version	Size	Page 13 Total 16
		REVO VERO	A3	VAP-SAN

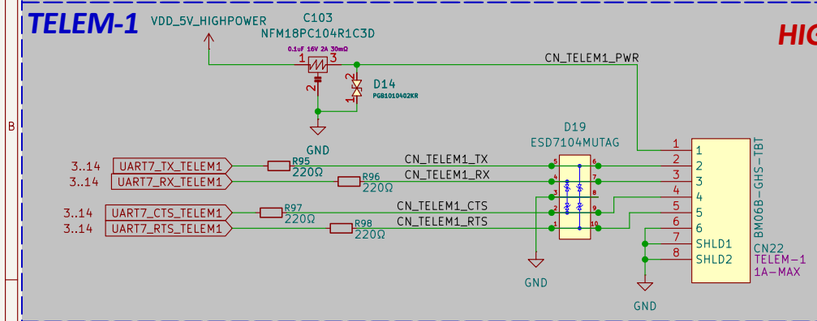
CAN-1

HIGH POWER



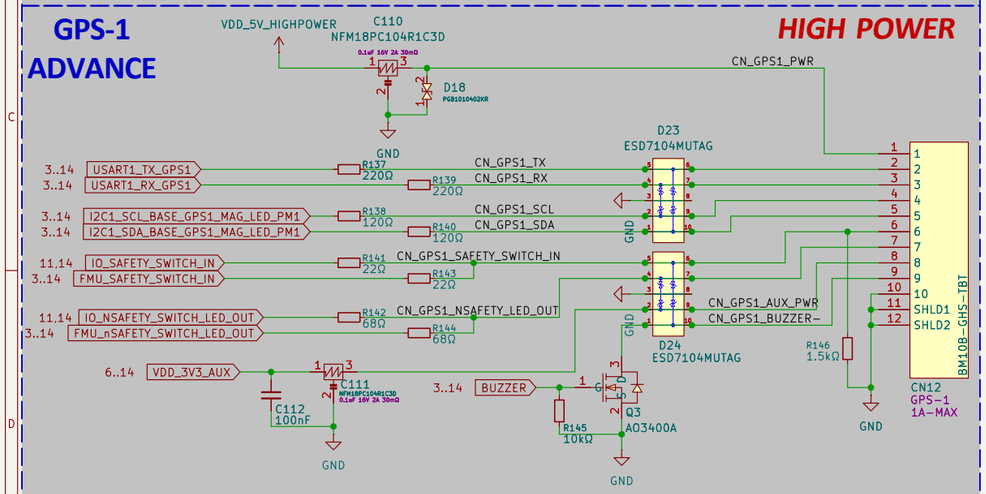
TELEM-1

HIGH POWER

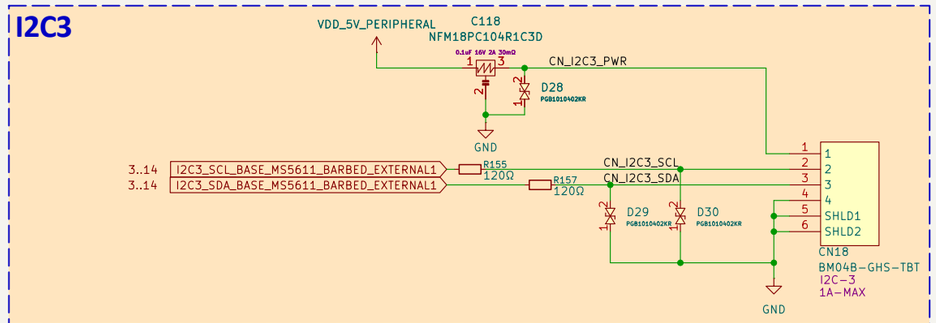


GPS-1 ADVANCE

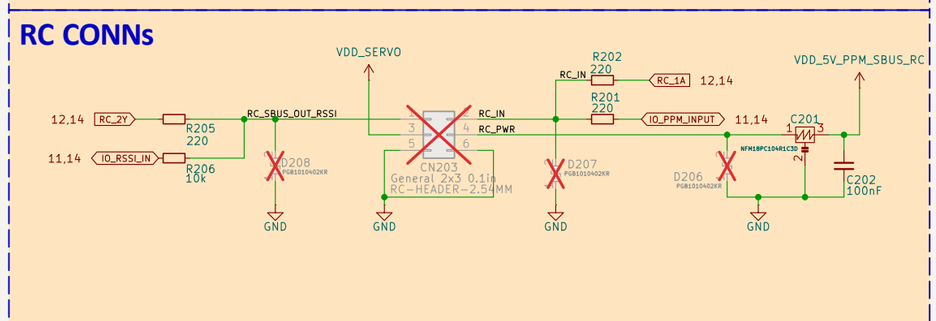
HIGH POWER



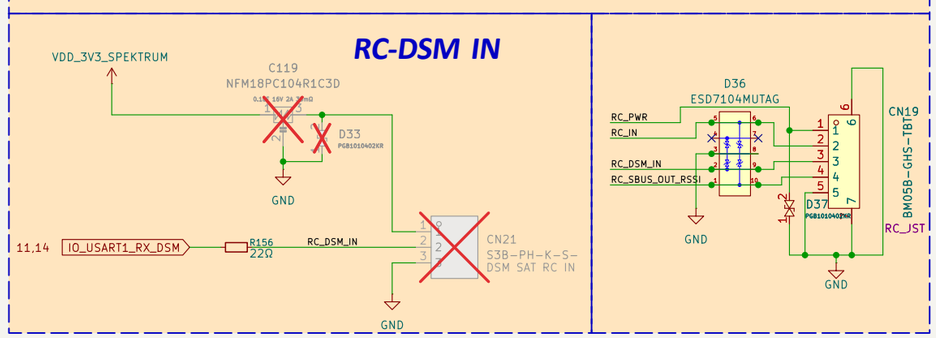
I2C3



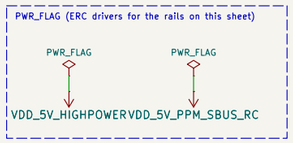
RC CONNs



RC-DSM IN

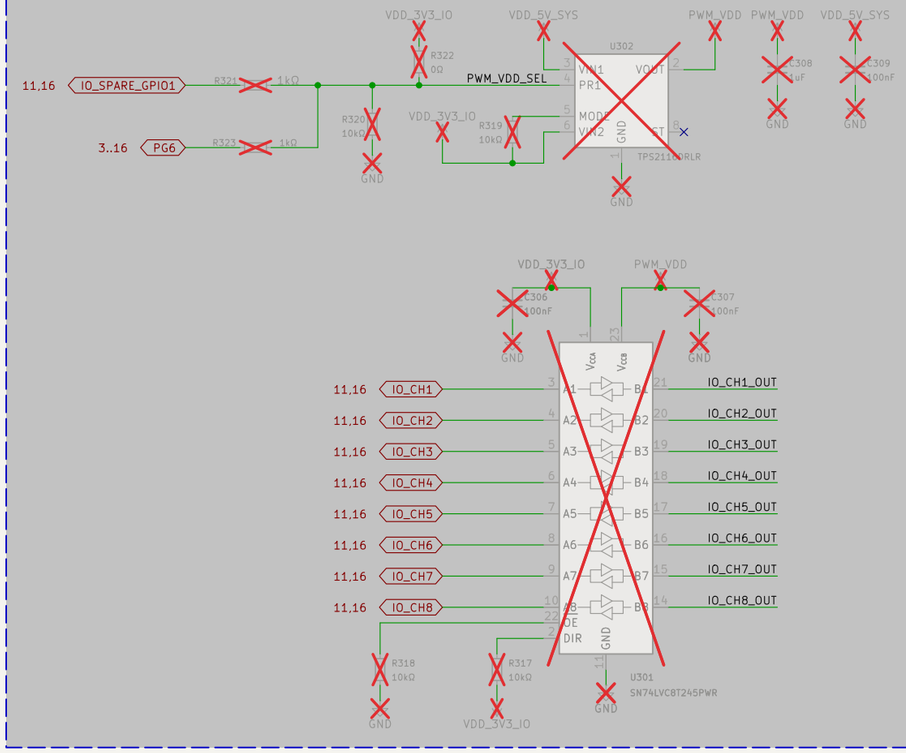


C4: CN203 (2x3 0.1 in header) stays DNP as an option.
R201 R202 R205 R206 C201 C202 D206–D208 are fitted: they also serve CN19 (GH RC IN).



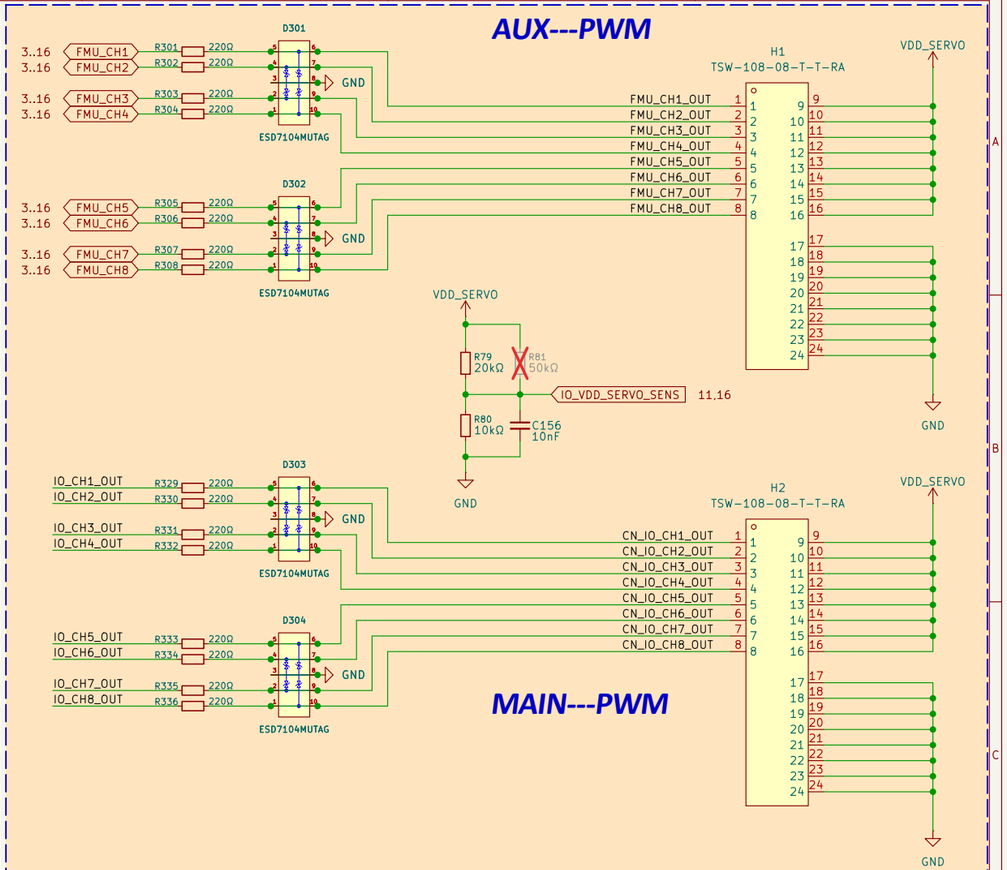
Schematic	**Naspier Core**	Create at	2026-09-20
Board	NASPIER CORE FLIGHT CONTROLLER SYSTEM	Update at	2026-10-29
Drawn	Ahmet Vapurcu	Page	**CARRIER_CONN_2**
Reviewed			
	Naspier Core		
	Version: REVO VERO	Size: A3	Page 14 Total 16
			VAP-SAN

AUX PWM VOLTAGE LEVET OPT.



FMU Drive with PG6	DIGITAL SET	PWM_VDD_SEL : 3.3V	MAIN PWM : 5V
IO Drive with SPARE_GPIO1	DIGITAL SET	PWM_VDD_SEL : 3.3V	MAIN PWM : 5V
R322 FITTED	ANALOG SET	PWM_VDD_SEL : 3.3V	MAIN PWM : 5V
OTHERWISE	ANALOG SET	PWM_VDD_SEL : 0V	MAIN PWM : 3.3V

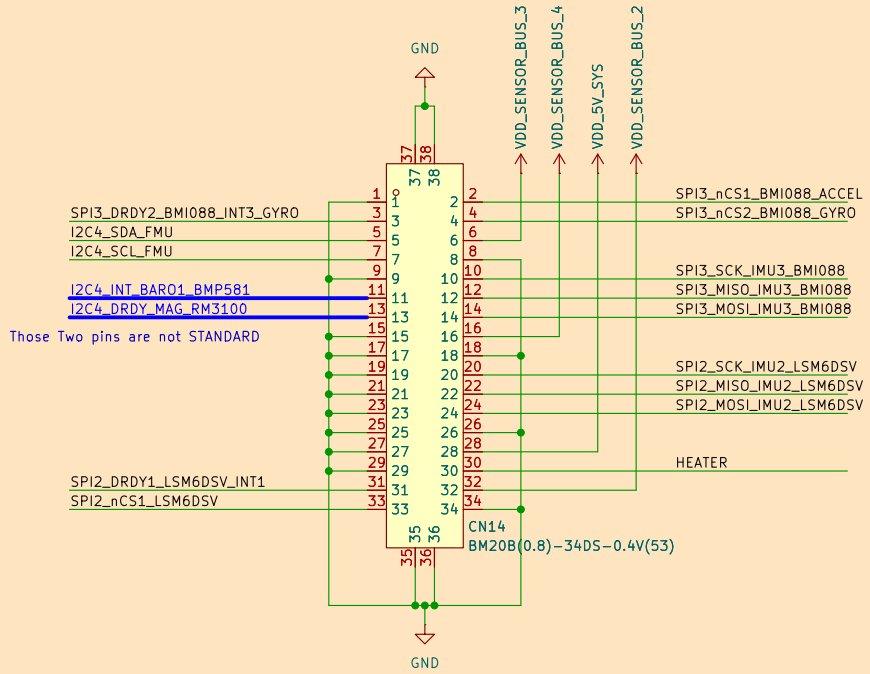
AUX---PWM

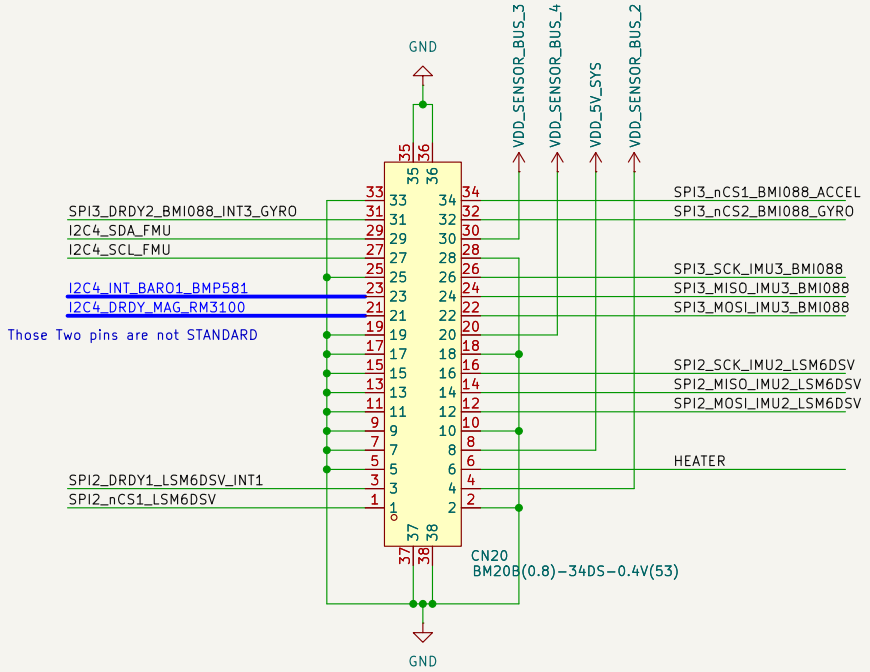


MAIN---PWM

Schematic	**Naspier Core**	Create at	2026-09-20
		Update at	2026-10-29
Board	NASPIER CORE FLIGHT CONTROLLER SYSTEM	Page	**CARRIER_PWM**
Drawn	Ahmet Vapurcu	**Naspier Core**	
Reviewed			
		Version	Size	Page 16 Total 16
		REV0 VER0	A3	VAP-SAN

PAB X3 34P SENSOR FFC CONNECTOR
NON-STANDARD CONNECTOR PINOUT





Schematic	**Naspier Core**	Create at	2026-09-20
		Update at	2026-10-29
Board	NASPIER CORE FLIGHT CONTROLLER SYSTEM	Page	**FLEX_CABLE**
Drawn	Ahmet Vapurcu	**Naspier Core**	
Reviewed			
		Version	Size	Page 15 Total 16
		REV0 VER0	A4	VAP-SAN